Labwork 2
Linear Regression

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1 Linear Regression

This section is about the implementation of Linear Regression from scratch.

1.1 Gradient Descent

Gradient Descent Algorithm to find the minimize value of the function $f(x)$:

$$x_{i+1} = x_i - lr \times f'(x)$$

1.2 Loss Function

The loss function is calculated by using mean squared error (MSE):

$$\text{MSE} = \frac{1}{2} \times \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

1.3 Implementation

Firstly, I initialize the parameters $w = [\mathbf{1.0, 0.0}]$. The model was running on 10000 epochs with learning rate 0.0001. I also applied the epsilon to stop training when the difference between two consecutive loss values is less than 0.01.

Epoch	Loss
1	1025.0266
2	396.4881
3	358.3248
4	353.8480
⋮	⋮
8019	46.1481
8020	46.1381

Table 1: Loss during training

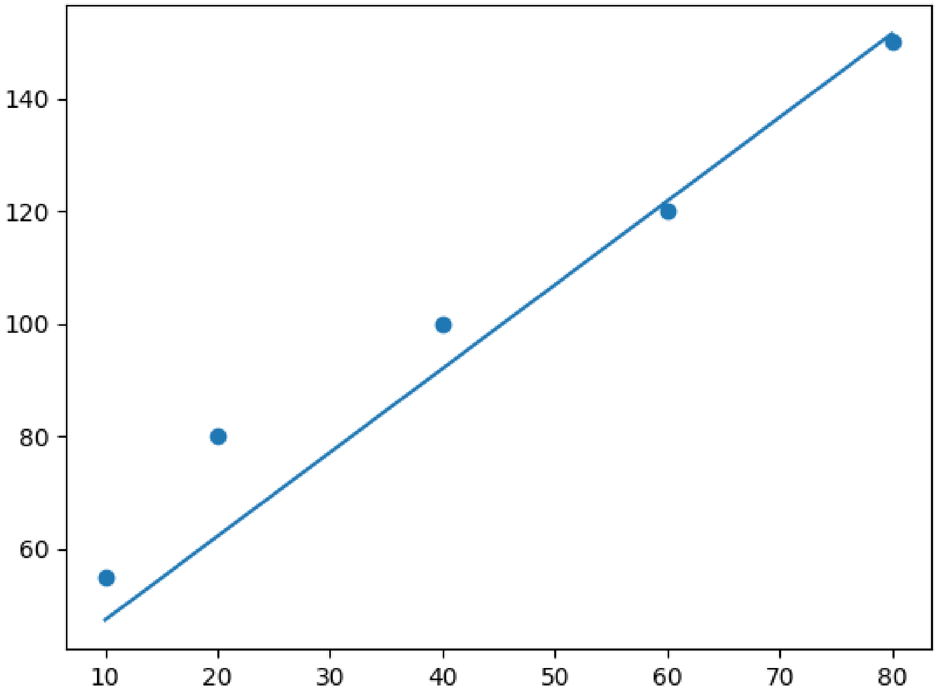

Figure 1: Prediction line